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The Impact of Dynamic Context Management on the Structural Quality and Functional Correctness of Code Generated by LLMs at the Repository Level: A Systematic Literature Review

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This systematic literature review aims to investigate the effects of dynamic context management on automatic code generation by language models, comparing it with static context strategies. The review seeks to identify how different forms of context management influence the quality of the generated code, considering structural metrics and the functional correctness of the software.

Planning

The overall objective is to systematically evaluate the methodological and empirical impact of dynamic context management on the structural quality and functional correctness of software generated by language models, comparing dynamic strategies with static context baselines and linear long-context ingestion.

The specific objectives are:

OBJ1: To identify primary literature that validates tools or algorithms for dynamic context management in the development and refactoring of agent-based software using LLMs.

OBJ2: Describe the experimental models, test environments, model configurations, and quality metrics used.

OBJ3: Compare the gains in structural metrics and functional success rates of dynamic context architectures with those of static comparators.

OBJ4: Synthesize the evidence, map discrepancies between RAG and long context windows, and propose methodological recommendations for quality assurance.

PICOC

- **Population:** Large Language Models for code generation
- **Intervention:** Dynamic context management
- **Comparison:** Static context strategies
- **Outcome:** Structural metrics and the functional correctness of the produced software
- **Context:** Software engineering projects at the repository level

Research Questions

1. RQ: How does dynamic context management in language models for code generation affect the structural metrics and functional correctness of the generated software compared to static context strategies?
2. SQ1: Does the performance of dynamic context strategies vary depending on the class of language model or the programming language involved?
3. SQ2: Which evaluation environments and testing infrastructures were used, and which demonstrated greater statistical robustness and reproducibility?
4. SQ3: Which dynamic context approaches and parameterizations maximized non-functional quality and functional correctness at the system level?
5. SQ4: Are the reported benefits confirmed in studies that control for data leaks, use temporal partitioning, random seeds, and open-source code?

Keywords and Synonyms

Keyword	Synonyms
Dynamic context management	RAG, adaptive context, adaptive context, agent memory, context engineering, context management, context pruning, context selection, dynamic retrieval, retrieval-augmented generation
Language models for code generation	LLM, code LLMs, language models, large language model
Static context strategies	baseline prompting, basic prompting, fixed context, large context window,

	limited context, long context, single-turn, static context, static prompting, static retrieval
Structural metrics and functional correctness of the generated software	code correctness, code metrics, code quality, functional correctness, maintainability, pass@k, software quality, static analysis, structural metrics, structural quality

Search String

("large language model" OR LLM OR "language models" OR "code LLMs") AND

("dynamic context" OR "dynamic retrieval" OR "context management" OR "context engineering" OR "context pruning" OR "context selection" OR "adaptive context" OR "retrieval-augmented generation" OR RAG OR "agent memory")

AND

("static context" OR "fixed context" OR "limited context" OR "basic prompting" OR "baseline prompting" OR "static prompting" OR "static retrieval" OR "long context" OR "large context window" OR "single-turn")

AND

("functional correctness" OR "code correctness" OR "pass@k" OR "code metrics" OR "structural metrics" OR "structural quality" OR "code quality" OR "software quality" OR "maintainability" OR "static analysis")

Sources

- ACM Digital Library (http://portal.acm.org)
- IEEE Digital Library (http://ieeexplore.ieee.org)
- ISI Web of Science (http://www.isiknowledge.com)
- Scopus (http://www.scopus.com)

Selection Criteria

Inclusion Criteria:

- IC1: Be a primary empirical study, published on or after 2024, written exclusively in English, and with full-text access available.
- IC2: Explicitly investigate language models or agent-based systems applied to the generation, refactoring, repair, modification, or active translation of source code at the repository level, involving multiple files
- IC3: Evaluate dynamic context management techniques in direct empirical comparison against at least one static context baseline.
- IC4: Evaluate the quality of the generated code using formal metrics of functional correctness or structural and non-functional quality.

Exclusion Criteria:

- EC1: A secondary or tertiary study, a purely conceptual article, an editorial, a short industry report, or an abstract without complete data, published before 2024, written in a language other than English
- EC2: Operate strictly at the level of a single function or isolated snippets; purely textual RAG without a software structure; general PLN without code; or passive static analysis without active modification of
- EC3: Not present a direct empirical comparison against a static reference context baseline or have context management only as a secondary element.
- EC4: Have the evaluation limited exclusively to information retrieval metrics without analyzing the generated code, or constitute a duplicate publication.
- EC5: Constitute duplicate publication; in this case, the most complete and recent version of the study will be retained.

Quality Assessment Checklist

Questions:

- QA1: Does the study provide source code, execution scripts, configuration files, data, prompts, random seeds, and API versions to enable replication of the experiment?
- QA2: Does the study clearly identify the models and static baselines, apply appropriate statistical tests to assess the significance of differences, and conduct experiments at the repository level or across multiple files?

- QA3: Does the study take measures to prevent data contamination or leakage and evaluate the stability and consistency of the generation across multiple runs, batches, or inference seeds?
- QA4: Does the study evaluate the functional correctness of the generated code through executable tests?
- QA5: Does the study analyze non-functional and structural attributes of the software or report metrics on computational efficiency, cost, or latency associated with dynamic context management?

Answers:

- Yes
- Partially
- No

Data Extraction Form

- [DQ1] DE1: Study identifier
- [DQ1] DE2: Authors
- [DQ1] DE3: Year of publication
- [DQ1] DE4: Country of the study
- [DQ1] DE5: Publication medium
- [DQ1] DE6: Publication type
- [DQ2] DE7: LLM model, version, size, and class
- [DQ2] DE8: Code granularity, languages, benchmark, and sample
- [DQ3] DE9: Dynamic context management technique
- [DQ3] DE10: Baseline context strategy
- [DQ4] DE11: Evaluation environment and infrastructure
- [DQ4] DE12: Experimental setup and feedback mechanisms
- [DQ5] DE13: Metrics and functional results
- [DQ5] DE14: Software Quality and Efficiency Metrics
- [DQ6] DE15: Reproducibility artifacts
- [DQ6] DE16: Methodological Rigor and Bias Control
- [DQ7] DE17: Stated limitations and threats to validity

Conducting

Digital Libraries Search Strings

ACM Digital Library:

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Scopus:

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Imported Studies

- **ACM Digital Library:** 456
- **IEEE Digital Library:** 7
- **ISI Web of Science:** 36
- **Scopus:** 106